3.7 Remote Debugging

Remote debugging will allow developers to attach external tools to a running Python process and execute code remotely, which can help in troubleshooting and development efficiency. The standard for the remote debugger connection is the Debug Adapter Protocol (DAP). Most popular IDE's (Integrated Development Environment IDE) support this protocol. The IDE offer to debug python code which is running on a remote target. The remote target runs a DAP-server which connects to a DAP-client, being the IDE. A remote debugger offers support for:

- Breakpoins
- Variables and expression watch
- Step in/out/resume

3.7.1 Non-Functional Requirements

NFR-ident	Non-Functional Requirement	Reference
NFR-RDB-001	The used IDE shall support the DAP (Debug Adapter Protocol) protocol to allow communication with the debugger.	
NFR-RDB-002	A DAP server shall be available on the target debugger	
NFR-RDB-003	When executing a python application, it shall use defined flags to enable and configure the remote debugging.	
NFR-RDB-004	The DAP-server shall only connect to the configured DAP-client endpoint, being the IDE.	
NFR-RDB-005	The DAP connection shall be reachable only through a local (private) network.	
NFR-RDB-006	When the DAP-client is not available the python application shall continue until connection is established.	